

Weekly Site Report

BLACK ROCK WEEKLY REPORT

Weekly PSD Report
EPIROC SA

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Weekly Service Report – Black Rock Mine Operations

Report type:	Weekly	Submitted to:	Sello Taku and Sipho
Week Date:	22 November 2025	End Date	27 November 2025
Reported By:	Philip Moller		
BEV:			
Service Tech:	Ivan Phiri (morning shift)	Service Tech:	Lukas Serumola (Night shift)
Service Tech:	Patricia Mazinyo (Afternoon shift) - Taking leave Thursday 04 Dec and Friday 05 Dec for a funeral.	Service Tech Assistant:	Patrick Yoko (Batteries)
Specialist:	Frikkie van der Walt (Batteries)		
Field Service:			
Service Tech:	Lucas Slaffa	Service Tech:	Mathews Mosemane
CAS Project:			
Team Leader:	Wisani Shilenge	Service Tech:	Phineas Sibanda
Service Tech:	Julius Manamela	Service Tech:	Joseph Mampa
CertiQ Project:			
Team Leader:	Segunda Fernando	Service Tech:	
Mobilaris:			
Team Leader:	Rupert Grobler		

Report Summary

General Summary

The purpose of this report is to give an overview of all the Epiroc activities at Blackrock Mine Operations (BRMO).

BEV - Critical Breakdown Summary

- DT171 – Ongoing breakdown. A-frame bearing damaged, bearing was replaced. With the removal and installation of the auxiliary motor oil and water entered some of the plugs and caused a wire to get damaged. Seeing insulation faults on machine. Boisy and Lukas from Epiroc to go on morning shift and assist to get machine back to production. One traction inverter changed and is functioning correctly. Busy loading software and parameters on the other traction inverter. Hydraulics overheating. Team continues with fault finding today.
- Continuing with audit on battery disconnects and compatibility. Photo report on connector conditions carried out and was shared with Sello and Piet.
- 3 Batteries on Breakdown. (VPY00011, VPX00031, VPX00026)

Outstanding audits:

DT162

FL113

CAS L9 Implementation - Critical Issues Summary

- 4 Machines left with software upgrades on Gloria.

Certiq & Mobilaris - Critical Issues Summary

- Machine availability to review the installation and collect evidence for the commissioning documents

Passport 360 Compliance

- BEV Technicians: **96%**
- CASL9 (Gloria): **100%**
- CASL9 (Nch2): **98%**
- CAS L9 (Nch3): **98%**
- Certiq & Mobilaris N2: **92%**
- Certiq & Mobilaris Gloria: **83% -**

BEV Report

Weekly machine utilization summary:

Saturday 22 November 2025:

1. FL 98 -
2. FL 99 -

3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 - Morning shift: Electrical breakdown – No feedback. (1H28min)
8. DT 147 -
9. DT 149 -
10. DT 150 -
11. DT 162 -
12. DT 163 -
13. DT 171 -

Sunday 23 November 2025:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 -
8. DT 147 -
9. DT 149 -
10. DT 150 -
11. DT 162 -
12. DT 163 - Morning shift: Electrical breakdown – No feedback. (6min)
Morning shift: 3rd party supplier fault – Strata. (1D01H05min)
13. DT 171 -

Monday 24 November 2025:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 - Night shift: Mechanical breakdown – BMS error – Recovered to battery bay and replaced VPY00041 with HVIL fault. (21H16min)
5. FL 112 -
6. FL 113 -
7. DT 146 - Afternoon shift: Electrical breakdown – Motor winding over heating fault. Found temp sensor cable loose. (2H)
8. DT 147 - Afternoon shift: Electrical breakdown – No feedback. (1H08min)
9. DT 149 - Morning shift: Electrical breakdown – Replaced M12 going from cab to aircon. (5H27min)
Afternoon shift: 3rd party supplier fault – Strata. (7H11min)
10. DT 150 - Night shift: Mechanical breakdown – Oil leak. Replaced hoses. (20H55min)
11. DT 162 - Afternoon shift: Electrical breakdown – 800V battery – Isolation fault on VPX00015, changed battery. (59min)
12. DT 163 -
13. DT 171 -

Tuesday 25 November 2025:

1. FL 98 -
2. FL 99 - Morning shift: Electrical breakdown – No feedback. (4H28min)
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 - Morning shift: 3rd party supplier fault – Petroman. (13min)
8. DT 147 -
9. DT 149 - Morning shift: 3rd party supplier fault – Strata. (1H28min)
Morning shift: 3rd party supplier fault – Strata. (1H50min)
10. DT 150 - Afternoon shift: Electrical breakdown – No feedback. (2H09min)
Night shift: Boilermaker breakdown – No feedback. (15min)
11. DT 162 - Morning shift: 3rd party supplier fault – Petroman. (1H15min)
12. DT 163 -
13. DT 171 -

Wednesday 26 November 2025:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 – Morning shift: Boilermaker breakdown – Half arrows. (3H59min)
Afternoon shift: Mechanical breakdown – H frame pin needs to be changed. (14H29min)
7. DT 146 - Morning shift: Electrical breakdown – Temp sensor, clean sensor plug. (5H33min)
8. DT 147 - Morning shift: Damages – light bracket damaged. (56min)
Morning shift: Mechanical breakdown – No feedback. (27min)
9. DT 149 - Morning shift: 3rd party supplier fault – Strata. (10H13min)
10. DT 150 -
11. DT 162 - Night shift: Electrical breakdown - DT162 recovered and replaced VPX-26 that had subpack #1 level 3 fault and had Isolation after wards and we saw that one of the clamps was too tight on HV plug going to the battery pack and Marko fixed it.
12. DT 163 -
13. DT 171 -

Thursday 27 November 2025:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 -
8. DT 147 -
9. DT 149 -

- 10. DT 150 -
- 11. DT 162 -
- 12. DT 163 -
- 13. DT 171 -

Re-occurring issues:

- 1. DT 146 – Issue with temp cable. Need to do proper fault finding on issue.

Improvement and actions:

- 1. DCDC campaign kicked off to reduce charger interruptions. (Ongoing)
4 batteries remaining to be converted this week.
- 2. Post 3,6 and 7 CCS connector cables need to be replaced. (Cause of High pin temp)
See below status of parts.

OPEN ORDER REPORT FOR BLACK ROCK: 12 NOVEMBER 2025

Order Date	Order Number	Item number	Description	Ordered	Shipped	Qty open	Status	ETA
2025/11/13	25001480/O7/MW	9464900654	GROMMET	3	0	3	Open	2025/12/05
2025/11/13	25001480/O7/MW	3180004087	SET,SPRING	3	0	3	Open	2025/12/05
2025/11/13	25001480/O7/MW	3180004088	HOLDER	3	0	3	Open	2025/12/05
2025/11/13	25001480/O7/MW	3180004092	KIT,SERVICE	3	0	3	Open	2025/12/09

- 3. Auxiliary motor spline grease TSNB not done. We need to schedule the machines to complete this campaign. Grease already on site.
- 4. Look into feasibility of morning shift Epiroc technician to oversee and assist on the electrical and mechanical services of the BEV's.
- 5. Connector audits on Machines almost done. 3 machines to be audited. (FL99, FL113 and DT162)
We will have to replace and send the changed-out cables to Epiroc to see if we can repair them, to save costs. (Discussed this with Kishore and he is aligned)
- 6. Change management need to be done on Regeneration knob.

SERVICES:

Nchwanning 3

Maintenance Schedule Des 2025				
Monday	Tuesday	Wednesday	Thursday	Friday
1-Dec-25	2-Dec-25	3-Dec-25	4-Dec-25	5-Dec-25
DT0118	DT0150	DT0131	DT0148	
FL0107	FL0080	FL0113		
HD0057	HD0058	HD0061	HD0062	HD0069

Battery and Charger Status Report

General summary

ST14 – B4 battery packs

- We currently have 10 x ST14 – B4 battery packs and 6 x ST14 machines operational underground. This is equal to the committed ratio of 1.6 batteries per machine.

MT42 – B5 battery packs

- We currently have 12 x MT42 – B5 battery packs operational underground. This is above the committed ratio of 1.6 batteries per machine.

160 kW Chargers

- Post 3,6 and 7 CCS connector cables need to be replaced. New 300A cables was order and received. Need to order upgraded spring set. Piet did order upgrade kit. Waiting parts.

General

- There are more noted charger stops; this is being investigated with initial indications being worn out charger and CCS battery inlet connectors. DCDC campaign to kick off 17 November 2025.

Table 1: Battery pack availability for week 2025

Battery Type	Battery ID	Status	Comment
B4 - ST14	VPY-00011	Breakdown	Battery is not charging. Busy with fault-finding.
B4 - ST14	VPY-00031	Working	
B4 - ST14	VPY-00051	Working	
B4 - ST14	VPY-00048	Working	
B4 - ST14	VPY-00049	Working	

B4 - ST14	VPY-00088	Working	Changed 2 subpacks
B4 - ST14	VPY-00086	Working	
B4 - ST14	VPY-00076	Working	
B4 - ST14	VPY-00083	Working	
B4 - ST14	VPY-00041	Working	
B5 - MT42	VPX-00016	Working	
B5 - MT42	VPX-00015	Working	
B5 - MT42	VPX-00023	Working	
B5 - MT42	VPX-00017	Working	
B5 - MT42	VPX-00010	Working	
B5 - MT42	VPX-00050	Working	
B5 - MT42	VPX-00031	Breakdown	VPC fault (4) - waiting feedback from Scania
B5 - MT42	VPX-00036	Working	
B5 - MT42	VPX-00028	Working	
B5 - MT42	VPX-00026	Breakdown	VPC safe state lvl 3. Waiting feedback from Scania
B5 - MT42	VPX-00024	Working	
B5 - MT42	VPX-00048	Working	
B5 - MT42	VPX-00044	Surface	On machine: 8997901500

Battery overheat summary:

VPX00028 – RCA to be done. Did not receive it yet.

FST Report Summary

Field Service on site:

CAS Report Summary

Report Type:	Weekly	Project Leader:	Hennie van Niekerk
Start Date:	15-Nov-2025	Report By:	Bradley Nast
End Date:	21-Nov-2025	Contract No:	BPE145D

Critical Issues

1. The team is to move to Gloria for software update and commissioning. Got new software. Will start installing this week.

Progress Update

1. All diesel loaders and a truck in NCH3 software is upgrade and commissioned on CAS L9.
2. All Loaders in NCH2 are Updated and live on L9.
3. All ST14 in Gloria need a software update and re commissioning. (4 machines left)

CertiQ & Mobilaris Report Summary

(To be reviewed with Vikesh and Rupert – Will be updated next week)

General summary

NCH 2 and Gloria are in the project phase. Installations are complete and commissioning is in progress. NCH3 is on a support contract, However the contract has lapsed.

NCH2:

25 machines scoped for the smart enablement project:

- 3 Machines are scrapped
- 2 Machines' commissioning document sent for signature by the Black Rock project team; those are currently not communicating.
- 12 Machines are complete, and Commissioning documents are signed.
- 1 machine work in a section with no Wi-Fi. Can't test and commission. Issue raised with Wayne
- 1 Machine not in use. Mine manager reserved future projects.
- 7 machines are under investigations.

Gloria:

13 machines scoped for the smart enablement project:

- 3 machines completed and commissioned.
- 8 machines under investigations.
- 2 machines, ECU not giving out date. It was report to Black Rock. Replacement is expected.

NCH3:

1. **Support contract expired.**

Support contract to be reviewed. Some S2 machines on Nchwaning 3 still struggling with connection issues even after software upgrades.

Segunda to assist.